Study Script - Outdoor Study

Black: script; Blue: comments (not read out); Red: actions

Read the Consent

Before we start, I will read the consent form to you.

You are invited to participate in a research study to explore how our AR system that augments important objects in the environment can help enhance your awareness of the surrounding environment. This study will begin with an initial interview on your demographic information, visual conditions, and experience with augmented reality. Then, we will introduce our augmented reality system. Next, you will take a walk outside with the system to evaluate the system. Lastly, we will have an exit interview on your preferences and suggestions for the system design. The entire study will last around one hour. If you feel tired or frustrated, feel free to tell me and take a break. You can take as many breaks as you want. We will provide a compensation of 25 dollars per hour and compensate your travel expenses for up to 30 dollars.

You will be audio- and video- recorded during the study. Audio and video recordings in their original form will only be accessible to the research team. Anonymized images or videos may be included in publications, academic presentations, and classrooms. The recordings will be retained indefinitely to be used in future research.

Now we will have your oral consent for the study. Would it be ok if we record the process?

1. **Do you have any questions about this research study?**
2. **Do you agree to participate in this study?**
3. **What is your full name? What is the date for today?**
4. **Do you agree to be quoted directly in publications without your name?**

Pre-study interview

First, I am going to ask about your demographic information and visual condition.

1. How old are you?
2. How would you identify your gender?
3. What is your visual condition?
    a. Are you Legally blind?
    b. What is your Diagnosis?
    c. What is your Visual acuity?
    d. What is your Field of view?
    e. How is your contrast sensitivity?

    f. How is your Color vision?
    g. Are you sensitive to light?
    h. What is your depth perception?
4. How long have you had this visual condition?
    a. Is this condition progressive or stable?
5. Are you using any tools to assist with your visual experience?
6. This study is based on Augmented Reality, or AR. It is a technology that overlays digital information onto the real world to create an interactive experience between digital and physical elements. Do you have any prior experience with Augmented Reality (Google Glass, Hololens, phone applications)?
    a. If yes, could you share the experience?

Tutorial

[Set up the tutorial scene: put objects on the table]
Now I am going to show you our system. This system will augment some objects around you to make them more visible. By augmenting surrounding objects, we want to help you be aware of the surrounding environment.

The system is based on this device called Hololens. It has a head-mounted display that you can see through. On the display there are two screens. Through the screens the system will display an augmentation on top of each object.

[Launch app before handing to participant]
Now, let's try the system. Would you like to put it on? There is a knob at the back, you can spin it to adjust the tightness. There is a strap on the top, you can adjust it to fit your head. Would you like me to help you put on the headset?

Can you see the table through the screen? Make sure you are seeing through the middle of the head-mounted display.
You can see there are some objects on the table, like knives, plates, bowls, cups, and so on. Let's first augment some of these objects.
[Visualize objects with yellow outline]
Are you seeing anything on the screen? Can you describe what you see? Is there a yellow outline around each object?

This is the system we developed. It can recognize objects in front of you, but we only augment important objects. Some objects that are not that important are simply not augmented. Also, as we will cover later, this system will augment objects differently based on their importance.

Instructions for outdoor exploration task

Task setup:

- walk behind the participant and video record everything in front. Another person walks with the participant to protect.
- Objects to augment:
    - Primary important: bicycle, skateboard, curb, pedestrian light, crossing lane & zebra stripe, train track
    - Secondary important: person, pole, fire hydrant, fence, utility box, traffic light, traffic sign, sidewalk

Route: 1330ft (400m); involved objects: pole, crosswalk, traffic light, sidewalk, train track, curb, person



For our task, we want to evaluate how effective our system is for walking outdoors. You will take a walk outdoors with the system. We will walk for around 10 minutes and cross the street twice. During the walk, please think aloud, talk about any thoughts you have about the system, such as your feelings about the augmentations, whether they are effective in helping you understand the outdoor environment, and so on. Safety first. You can walk at your comfortable pace. We will not be timing your walking speed or time. We will also walk with you to ensure safety.
Do you have any questions?

First, let's find the design that works best for you outdoors.
[Demo session: outside southern entrance to the CS building]

Please put on the headset. Are you seeing any augmentation on the pedestrians, the pole, and the curb?

Discuss augmentation design: Let's first talk about augmentation design. The yellow outline is one design, but we also have other designs. We have three design options. Let's go through each of them and discuss your preference.

- Outline: our first design is a bright yellow outline of each object. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?
    - If complain about outline color: let's adjust the color. Which color works better for you? How about now?
    - If complain about outline thickness: let's adjust the thickness. Is the outline too thick or too thin? How about now?
- Solid: the second design is a solid overlay on top of each object. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?
    - If complain about overlay color: let's adjust the color. Which color works better for you? How about now?
- Icon label with pointer: the third design is an icon label that hovers above each object and has a line connecting to the object. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?
    - If complain about icon color: let's adjust the color. Which color works best for you? How about now?
    - If complain about icon size: let's change the icon size. Is it too big or too small? How about now?

Okay. We've seen and customized all three design options. Now, let's go through them again, and pick the one you like the most.
[show all augmentation design options with the best customization]
Among the three options, which one works best for you?

Discuss distinguishing objects: We've discussed different augmentation designs, but so far, we assumed all objects are equally important and we augment them with the same design. However, there could be some objects that are more important than others. For example, there might be certain things that are more important and you want to see right away, versus others that are secondary important, so still important but just good to know. We want to use different designs to distinguish object importance so you can capture the most important objects immediately.

We have different designs to distinguish objects. Let's go through each of them and discuss whether you like it or not. Let's suppose the curb and bicycle are more important.

- Yellow / blue: the first design is to use a bright yellow outline for the more important objects and a blue outline for the others. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?
    - (If complain about specific colors): which color would you want for the "more important" objects? What about the "less important" ones? Let's try XX. How about now?
- Outline + icon / outline: the second design is to use an outline for all objects. For the important objects, we add an icon to each of them. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?
- Solid / outline: the last design is to use a solid overlay for the more important objects, and an outline for the others. What do you think of this design? What aspects do you like about this design? What do you dislike about this design?

Okay, we've seen and customized three ways of distinguishing between objects. Let's go through them again and pick the one you like the most.
[show all ways of distinguishing objects with the best customization]
Among these three designs, which one works best for you?

Now, you can freely explore the system outdoors to familiarize yourself with it. Then, we will start the task.

Now, let's take a walk with the system. Remember to think aloud and talk about your feelings and thoughts about the system, such as whether they are effective in helping you understand the outdoor environment.

Probe one object: did you see an augmentation for XXX? What do you think of this augmentation? Did it help you see the object?

Thank you! Let's walk back to the lab to finish the last part of the study.

Exit Questions

Before we finish the study, we have some exit questions on your experiences with the system and the augmentations.

- What do you think of the system? Do you like it? Why?

- In your opinion, what are the pros and cons of augmenting objects versus not augmenting them?
- In your opinion, what are the pros and cons of augmenting different objects differently versus in the same way?
- Do you have any other design ideas for the augmentations to support your awareness of the surrounding environment?

Thank you. That is the end of the study.